The Relational Knot

Why the Most Intractable Problems Across the Sciences Defy Explanation

Introduction: The Substance Trap

Across the entire landscape of human knowledge — from physics to philosophy, from biology to political science — there are problems that have resisted generations of our best minds. We have gathered immense data, built sophisticated models, and developed ingenious experimental methods. Yet the deepest puzzles remain stubbornly unresolved. Why?

A close look reveals something striking: these problems share a hidden architecture. They are not simply "hard questions" about weird objects or complex systems. They are all *relational* in a sense that our standard ways of thinking cannot capture. Our intellectual inheritance — the metaphysics we breathe without noticing — treats the world as composed of basic substances that have intrinsic properties and then enter into secondary relations. In this picture, things come first; how they connect comes later. A particle has mass and charge *before* it interacts. A gene has a function *before* it acts in a cell. A mind has beliefs and desires *before* it enters a social context. A fact is a fact, a value is a value, and the bridge between them is a separate, external operation.

The problems listed below defy resolution because they are *constitutively relational*. In each case, the relation is not an afterthought — it is what constitutes the relata in the first place. The parts are not prior to the whole, the observer is not separable from the observed, cause and effect are not independent chains, pattern and substrate co-define each other, and normative and descriptive dimensions are intertwined from the start. When we try to study such phenomena with a substance-first ontology, we systematically sever the very relations we need to understand. The result is not a temporary impasse but a conceptual impossibility.

What follows is a detailed survey of these intractable problems, organized by the relational archetype they embody. Each section names the problems, states clearly why they resist explanation, and exposes the relational nature that a substance-based framework cannot accommodate.

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1. Emergence Without a Stage: The Part–Whole Relation Reversed

Problems

- Quantum gravity (physics): Reconciling general relativity with quantum field theory to describe spacetime at the Planck scale.
- The binding problem (neuroscience): How the brain integrates separately processed features (color, motion, shape) into a single, unified conscious percept.
- Genotype–phenotype map (biology): How a one-dimensional genetic sequence yields the four-dimensional developing organism across scales — from protein folding to tissue organization.
- Micro–macro aggregation (economics): How individual choices and market interactions give rise to macroeconomic phenomena like business cycles, inequality, and secular stagnation.
- Water's anomalies (chemistry): The molecular origin of water's density maximum, high heat capacity, and supercooled behavior — emergent from a simple H_2O molecule.
- Personality stability vs. change (psychology): How core personality traits emerge from dynamic interactions of genes, environment, and life history, and whether they can be deliberately altered.

Why They Defy Explanation

In a substance ontology, we assume that a whole is the sum of its parts and their interactions. The strategy is reduction: find the smallest units, characterize their intrinsic properties, then compute how they assemble into larger structures. This works brilliantly for many systems. But in the problems above, the whole is not merely a sum; it imposes *organizing constraints* that reach back down and redefine what the parts themselves are.

In quantum gravity, spacetime itself is not a background stage but a relational network that emerges from quantum entanglement. We cannot start with "gravitons" as little particles in a pre-existing spacetime because spacetime is what their relations *produce*. In the binding problem, a unified conscious percept is not an extra step added after feature detection; the features are already shaped by top-down integrative processes, so a color is never "just" a color but is always already part of a scene. The genotype–phenotype map is not a linear translation because the developing organism is a system of mutual constraints — gene expression depends on tissue context, mechanical forces, and environmental inputs that cannot be factored out. Macroeconomic crises are not simply large numbers of individual errors; the idea of an "individual preference" is itself shaped by institutional rules, social norms, and the very market dynamics it supposedly explains.

In every case, the *relation of part to whole is reciprocal and constitutive*. The part has the nature it does only by virtue of its role in the whole, and the whole arises from the coordinated interaction of parts that it itself organizes. Trying to build up from independent parts assumes a separation that does not exist at the fundamental level. We lack a logical framework that treats the whole-parts relation as primitive rather than composite, and so we remain stuck.

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2. The Knower in the Known: Self-Referential Entanglement

Problems

- The measurement problem (physics): How a definite outcome arises from quantum superpositions, and where the "cut" between observer and observed lies.
- The hard problem of consciousness (philosophy, neuroscience): Why and how physical processing in the brain yields first-person subjective experience.
- Scepticism about the external world (philosophy): Whether we can justifiably know that an external world exists independently of our perceptions.
- Free will and the Libet experiments (neuroscience, philosophy): Whether neural readiness potentials preceding conscious intention mean that our sense of volition is an illusion, and what truly initiates action.
- Rational expectations and reflexivity (economics): How beliefs about markets change the markets themselves, making predictions inherently self-altering.
- Replicability crisis (psychology): The extent to which published findings reflect genuine effects versus experimenter degrees of freedom and social incentives.

Why They Defy Explanation

Standard science imagines a clear separation: there is an object of study *out there*, and an observer *in here*. The observer uses instruments and methods to measure the object's properties without fundamentally altering what is measured. This separation is precisely what fails in these problems.

In quantum mechanics, the measurement apparatus and the system form an indivisible whole. Asking "what is the electron's spin before measurement?" is like asking "what is the left side of a relation without the right side?" — the property is not possessed intrinsically but only in the context of a measurement interaction. The observer cannot be subtracted out. The hard problem of consciousness arises because we try to locate experience as a *property inside a brain*, when experience is better understood as a *relational process between an embodied organism and its world*. We look for the “feeling” as a substance when it is a mode of being-in-relation. Free will debates become intractable when we assume a linear causal chain from brain states to actions, ignoring that the “self” that decides is not a separate substance causing neuronal events but the integrated form of a living system in action.

In economics, rational expectations theory acknowledged that agents’ models of the economy affect the economy, but it still assumed a convergent truth “out there.” Reflexivity — the two-way shaping between perception and reality — breaks that convergence, making the market fundamentally unforecastable from an external stance. The replicability crisis in psychology is not merely a statistical problem; it reflects the fact that the scientist’s choices (which measures, which analyses) are part of the very phenomenon being studied, and no purely methodological fix can remove the investigator’s participation in constituting the evidence.

Across these cases, the knowing relation precedes and shapes the knower and the known. We cannot step outside the relation to view it from nowhere. Our substance ontology insists on a God’s-eye view that is unavailable in principle.

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3. Causal Webs, Not Causal Chains: Constitutive Interdependence

Problems

- Microbiome–host causality (biology): Moving from correlation to mechanism in understanding how gut microbes influence mood, immunity, and disease — and vice versa.
- Missing heritability (biology): Why identified genetic variants typically explain only a small fraction of the heritability of complex traits, because gene–environment interactions are not additive.
- Causes of war (political science): Why states end up in costly, ex-post irrational wars despite rational-choice theories.
- Political polarisation (political science): What drives affective and ideological polarisation, and whether reforms can reverse it.
- Predicting financial crises (economics): Why asset bubbles and systemic risk remain elusive despite vast monitoring.
- Ecosystem tipping points (biology): Predicting when a complex ecosystem will irreversibly shift to a degraded state.

Why They Defy Explanation

We habitually think of causation as a sequence: A pushes B, B pushes C. Controlled experiments try to isolate A's effect by holding everything else constant. But in the problems above, causal factors are mutually constitutive — they do not act on each other as independent levers; they co-evolve, form feedback loops, and define each other's meaning.

The microbiome–host relationship is not a one-way street. Gut bacteria influence the brain, and the brain influences gut motility and immune response, creating a systemic loop where "cause" and "effect" are only momentary labels. Missing heritability is not a gap waiting to be filled by more genes; it reflects the fact that the phenotype is built by a *developmental relation* between genetic variation and environmental context. Twin studies assume you can partition variance into independent sources, but the gene–environment interaction is not a sum of separate contributions; it is a dynamic product. Political polarisation is not caused by one side's properties but by a relational dynamic of identity-by-opposition; each side's stance is partially defined by its negation of the other, so there is no anchor point of "true" preference before the conflict. Financial crises arise from the interplay of risk models, investor behaviour, and regulatory responses — each factor is itself shaped by the anticipation of what others will do.

In every case, the causal relation is a field of mutual dependence rather than a chain of separate billiard balls. The standard methods of causal inference — hold other factors constant, randomize, assign independent effects — break down when factors are not independent by nature. The world here is *relational all the way down*.

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4. Thresholds of Being: The Emergence of New Relational Forms

Problems

- Origin of life (biology, chemistry): The transition from abiotic chemistry to the first self-replicating, evolving entities.
- Origin of the universe (physics): What set the initial conditions, and what, if anything, came before the Big Bang.
- Major evolutionary transitions (biology): The origins of eukaryotes, sex, multicellularity, and eusociality — leaps in organizational complexity.
- Matter–antimatter asymmetry (physics): Why the early universe produced a tiny excess of matter over antimatter.
- Emergence of consciousness (neuroscience, psychology): How subjective experience arises in the developing infant brain.

- Function of non-coding DNA (biology): How vast stretches of the genome that don't code for proteins nevertheless enable the regulatory complexity of life.

Why They Defy Explanation

An origin is not simply a change from one state to another. It is the coming-into-being of a new form of relation that has no precedent in its predecessor terms. Before life, there is no "self" to distinguish from "environment." Before the universe, there is no spacetime container in which events occur. Before multicellularity, there is no collective entity with its own fitness interests.

Our standard causal explanations require a backdrop of pre-existing entities that interact and produce a new arrangement. But at a genuine origin, the new relational pattern *constitutes the entities it relates to*. The first living system was not a new substance added to chemicals but a novel *self-organizing relational closure* — a network of reactions that produced and maintained its own boundary. Asking "what caused life?" assumes a cause and an effect that already exist as separate things, but the very distinction between system and environment is what life brought into being. Similarly, the Big Bang is not an explosion in a pre-existing space; it is the limit where the relational fabric of spacetime itself becomes undefined. The matter–antimatter asymmetry requires an interaction that treats matter and antimatter differently, but that asymmetry is a *relation of difference* that cannot be explained from a perfectly symmetric starting point without an external differentiating factor — which there is, by definition, no room for.

In biology, the function of non-coding DNA is a classic case. We once called it "junk" because we imagined DNA as a linear code for proteins, with genes as distinct substances. But regulatory networks, epigenetic marks, and three-dimensional chromosomal architecture reveal a system in which the "meaning" of any sequence depends on its relational context. The function is not in the sequence but in the dynamic pattern of interactions. The origin of such regulatory complexity is not a list of added genes but a threshold of a new kind of relational information system.

These problems resist solution because we cannot use the very categories that the transition brought into existence to describe what came before. A relations-first ontology would acknowledge that the new relation is irreducible to its antecedent components, because those components did not exist as such before the threshold.

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5. The Is–Ought Relation: Value as Relational Affordance

Problems

- Optimal discounting for climate change (economics): What rate should society use to trade off present costs against future benefits, and how do we handle intergenerational equity?
- Moral realism vs. anti-realism (philosophy): Are there objective moral facts, and if so, what is their nature?
- Taxonomy of mental disorders (psychology, psychiatry): Are disorders like depression discrete natural kinds or arbitrary thresholds on continuous spectra?
- Technocracy vs. populism (political science): How can democracies navigate the tension between expert knowledge and popular sovereignty?
- Arrow’s impossibility theorem (political science, economics): How to aggregate diverse individual preferences into a coherent social choice without violating minimal fairness conditions.
- Meaning of life (philosophy): Does life have an objective purpose, or must meaning be subjectively constructed?

Why They Defy Explanation

A substance ontology splits the world into facts (what is) and values (what ought to be). Facts are discovered by science; values are either subjective preferences or objective moral truths of a different substance type. The bridge between them becomes a mystery, and many problems then appear as unbridgeable chasms.

But these problems are intractable precisely because values are not separate substances added onto facts; they arise from the relational structure of agency in a world. The discount rate for climate policy is not a purely technical number that economists can uncover, nor is it a purely ethical axiom. It embodies a *relation between present and future persons* — a question of what we owe to those whose existence depends on our choices. The number is a quantitative expression of a form of intergenerational solidarity. Treating it as a fact without value or a value without fact misses that it is a *relational quality* of our collective life.

Moral realism debates oscillate endlessly because both sides assume that moral properties, if real, must be either natural properties (like “pleasure”) or non-natural, queer entities. A relational view sees moral status as something that emerges in the encounter between vulnerable beings and agents capable of response. It is neither a property of actions alone nor a projection of the mind, but a *real relation of obligation* that is as much a part of the world as causation. The taxonomy of mental disorders fails when we insist that a condition must be either a natural kind with a biological essence or a social construction. In practice, “disorder” is a relational judgment involving harm, dysfunction, and social norms — the boundary is and ought simultaneously.

Arrow's theorem shows that any voting rule that respects certain minimal fairness conditions must sometimes produce irrational collective outcomes. The intractability is not a bug; it reveals that "the will of the people" is not a substance that exists prior to the aggregation method. The collective decision is a *relational artifact* of the procedure and the preferences — you cannot make the relation transparent because it is constitutive. Technocracy vs. populism is the same tension: expertise and democratic will are not two substances to be balanced but two moments in the relational process of collective self-governance, each partly defining the other.

Each of these problems persists because we seek to ground the normative in either a fact or a pure value-substance, when in reality the normative is a dimension of the relation itself.

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6. Pattern and Medium: The Informational Relation

Problems

- The neural code (neuroscience): In what format (rate, temporal, population code) is information represented and transmitted in the brain?
- Memory engrams (neuroscience): What is the physical basis of a specific memory, and how are memories consolidated, transformed, and retrieved?
- Black hole information paradox (physics): Is information destroyed when something falls into a black hole, and how is unitarity preserved?
- Function of non-coding DNA (biology): What fraction is functionally relevant, and how does it regulate genome biology beyond coding sequences?
- The species problem (biology): Whether "species" correspond to real, discrete entities in nature, and what is the best unified species concept.
- Personal identity over time (philosophy): What makes a person at time t_2 the same person as at t_1 — psychological continuity, bodily continuity, or something else?

Why They Defy Explanation

Information, we intuitively think, is a *thing* that can be stored, transferred, or lost. A memory is a trace in the brain. A gene carries a code. A black hole swallows matter and might destroy the information it carried. This is the substance view of information — a message that travels through a medium like a letter in an envelope.

But these problems reveal that information is not a substance but a relation between a pattern and its interpretive context. The neural code cannot be found by looking at spike trains alone, because a spike train means nothing without the downstream reader's expectations and the organism's ongoing activity. The "code" is the *predictive relationship* between firing patterns and the behavior or perception they enable — it is not in the spikes but in the system of mutual attunement. The memory engram is not a fixed physical trace; memory is a reconstructive process where the same neuronal networks are continuously reshaped by present context. The past is not stored as a file; it is re-enacted in a relational dynamic.

The black hole information paradox arises because we imagine "information" as an entity that goes into the hole. But if information is a correlation structure between systems, then when one system is causally disconnected behind an event horizon, the correlations are not destroyed — they are *relocated* in subtle entanglements with the Hawking radiation. The paradox is a clash between our substance-view of information and the relational nature of quantum states. The function of non-coding DNA is not a property of a sequence but of its dynamic interaction with transcription factors, chromatin state, and cellular context. The exact same sequence can be functional or not depending on the relation, so we cannot assign "function" as a predicate. The species problem is the same: organisms are related by descent in a branching, reticulating network. A "species" is not a substance with an essence but a relationally defined lineage segment. Unified species concepts fail because we want a thing when reality is a nexus of relations.

Personal identity resists resolution because we look for a substantial self that persists — a soul, a brain region, a core narrative. But the self is an ongoing relational process: a history of interactions, a narrative that weaves past and future, an embodied engagement with a world. The continuity is not a substance that endures but a *pattern of relating* that maintains itself.

Conclusion: The Need for a Relations-First Ontology

What unites every problem described above is not a lack of data or clever models. It is a fundamental mismatch between the nature of the phenomenon and the conceptual grammar we bring to it. That grammar — our default substance-property ontology — privileges:

- Entities before interactions
- Intrinsic properties before relational roles
- Independent causes before mutual dependencies
- Observer–observed separation before participatory entanglement
- Facts before values
- Pattern before medium

The most intractable problems across the sciences are those where this ordering must be reversed. The relation is not what happens after entities are already given; the relation is what gives entities their identity and reality in the first place. We do not yet have a well-developed, mathematically rigorous, and experimentally operationalized *relations-first ontology*. We have glimpses — in process philosophy, in enactive cognitive science, in relational quantum mechanics, in network theory, in developmental systems biology — but no unified framework that can be applied routinely.

These problems have resisted explanation because we have been trying to solve them with tools that dissolve the very thing we are trying to understand. To make progress, we will need not just new theories within the old metaphysics, but a new metaphysics altogether — one that places relation, process, and co-constitution at the foundation of what it means to be.